

Section Heat Exchanger Upgrade

1.2.47.1 Rev. 0 (formerly CLS 2.1.67)

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REVISION HISTORY

<i>Revision</i>	<i>Date</i>	<i>Description</i>	<i>Author</i>
0	2000-June-27	Issued	N.G. Johnson

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1. INTRODUCTION

Maintaining ± 0.1 °C temperature regulation on the section heat exchangers has been an ongoing difficulty during SAL operation. In part this is because the heat exchangers had two functions. The sections had to be heated to the correct temperature when the RF was off. When RF power was being delivered to the section the excess heat had to be removed. The transient loading as RF was switched on or off tended to cause temperature fluctuations which took several minutes to stabilize. The air activated valves would regularly stick, resulting in jerky operation which caused small oscillations around the desired temperature. The PID temperature controllers required regular calibration and had to be operated in a temperature regulated cabinet to work with thermocouple sensors which were required to withstand the radiation levels near the linac. Relatively little information was available for diagnostic monitoring and logging.

The proposed CLS operation repetition rate of 1 Hz reduces the radiation levels in the linac and the heating caused by the RF. These differences offer a potential for improved temperature regulation, although the heat exchanger control system will need to be modified. Diagnostic monitoring would be added as part of these modifications.

This report outlines the proposed changes to the section heat exchanger system. The following heat exchangers will be modified:

- Linac section heat exchangers (8)
- Low Energy
- ECS
- BTL (renamed Linac to Booster, LTB1)

The injection/extraction heat exchanger will be removed, its magnets will be moved to the LTB1 heat exchanger. This project does not include relocating these magnet connections or plumbing to new magnets in the LTB line.

This project does not include maintenance and repairs on faulty equipment, such as valves, which may be discovered during the course of this work.

2. SCOPE OF WORK

2.1 Replacing Type J Thermocouple with Platinum RTD

A junction of two different metals produces a voltage potential which varies with temperature. This effect is the basis for thermocouple based temperature measurement. However, there are additional junctions formed when the thermocouple conductors are attached to the measuring device. Variations in temperature at the measuring device produce voltage variations which can be misinterpreted as temperature/voltage variations at the sensor. The controller can implement circuitry to compensate for variations in ambient temperature, but this proved to be insufficient. Regulating the temperature of the cabinet the sensors were mounted in did reduce this effect.

RTD based measurements are based on the variation of a sensor's resistance with temperature. Thermocouple effects do still occur, but are a much smaller effect. A second problem is self heating from the measurement current through the RTD. Since the devices are submersed in water for this application self heating will not be significant. Sensors will be ordered with 100 Ω resistance and European temperature coefficients.

SAL had tried using RTD sensors in the heat exchanger controls, but radiation levels were high enough to damage the insulation and the sensor lifetime was short (1-2 years). With 1 Hz operation, beam pulse lengths of 130 nS, and typical operation of a few hours/day the integrated radiation dose will be much lower than during SAL operation. Sensor lifetimes should be correspondingly longer.

RTD sensors are available in the same size as the existing thermocouple probes, so the mounts in the existing sections can be used. The RTD will replace the type J probe at the water supply connection to the section. The existing type J thermocouples will be used for diagnostic monitoring. The thermocouple which was replaced will be relocated measure the heater inlet water temperature.

2.2 Eliminate Heat Exchanger and 3 Way Valve

The three-way valve has been the most unreliable component in the system. The valve would continuously operate over a very small range. Buildups would develop at the end of the small movements, and cause the valve to stick, operating in a jerky manner. This sticking, then releasing caused corresponding temperature fluctuations, which could be stopped by operating the valve through its entire range for several cycles.

With the proposed 1 Hz operation of the CLS linac, RF heating of the sections is negligible. The klystrons produce approximately 25 MW of RF, for 1.5 μ S, or an average of 40 W of heating. Since the sections are operating between 40 and 45 °C this power can be dissipated to the ambient air. (The EROS storage ring magnets will not be powered, so the temperature in the linac hall should remain cool, but an additional fan coil cooling unit can be added if the ambient temperature becomes too warm during summer months.) The three way valve and associated heat exchanger element can be removed. SCR voltage regulators, with analog 4 – 20 mA inputs, are already installed on the section heaters. These regulators will be replaced with solid state relays, driven with pulse width modulation outputs from the programmable logic controller.

Sections 0, 1, 5, 6, and 7 have the heater and 3 way valve located close to the section. After the valve is removed the heaters will be relocated near the pumps and heater control. This will require some plumbing materials.

Some of the removed plate to plate heat exchanger elements will be used to upgrade old the old low energy and ECS heat exchangers.

2.3 Switch to PLC based control system

A programmable logic controller will be used to implement the PID control loops. The PLC will also be used to implement the interlocks and on/off control. This will allow the EPICS control system to monitor the control system and interlock status. One CPU should be sufficient for all eleven heat exchangers.

The PLC will have the following interlocks/inputs:

- Reservoir level
- Overtemperature
- Waterflow
- On/Off
- Section input temperature (RTD)

and the following outputs:

- Heater Power (PWM)

- Pump motor contactor
- Heater contactor

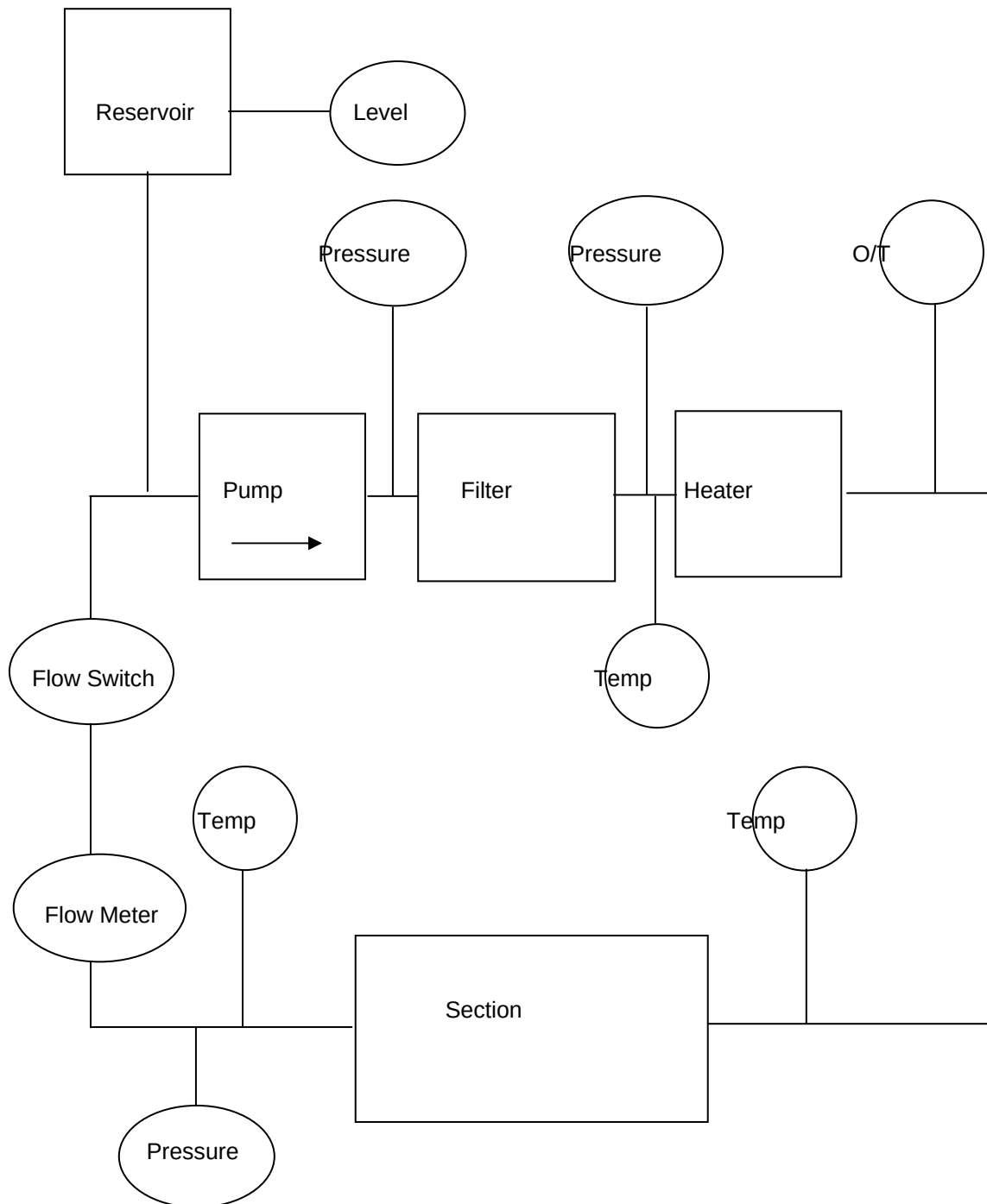


Figure 1: Section heat exchanger sensor locations

2.4 Additional Diagnostics

Several additional parameters can be monitored for diagnostic purposes. These could be connected to the PLC, or to a Fieldpoint based system. Monitoring the signals with the PLC

would allow them to be used as interlocks, but is more expensive . These signals are not required for interlocks, so they will be monitored by the Fieldpoint devices.

- These signals will include:
- Section output temperature
- Heater input temperature
- Section output pressure
- Pump output pressure
- Filter output temperature
- Water flow rate

The sensors will be configured as indicated in the block diagram. The two flow sensors may be located differently, but since this is a series system this will have little effect.

3. TEST RESULTS

One of the sections was modified to use the variable heater form of regulation. The three way valve was driven to the position which bypassed the heat exchanger element. The existing thermocouple input was used for temperature feedback. A second thermocouple was connected to a chart recorder. The ambient temperature in the linac hall was 22 °C, and the section temperature was set to 46.8 °C On the controller, the temperature appeared to be well regulated, but as the temperature of the room varied the chart recorder showed that the section temperature varied also. The reference thermocouple was replaced with an RTD element and the test repeated. Over a six hour period the temperature variation was +/- 0.05 °C

4. BUDGET

Three scenarios for the upgraded control system are presented in Table 1. A minimal PLC system with no additional diagnostics would cost \$18,400. The new diagnostics, using Fieldpoint data acquisition, would cost \$24,700 more than the minimal system. The control systems budget has been updated to include these costs.

Table 1: Budget		PLC Sensor	PLC I/O	Minimal System	Field point Sensor	Field point I/O	Fieldpoint addition to minimal system
Temperature Inputs		RTD Probe			Existing Type J Probe		
	Section Input	\$186.00	\$277.25	\$463.25			
	Section Output	\$186.00	\$277.25			\$89.38	\$89.38
	Heater Input	\$186.00	\$277.25			\$89.38	\$89.38
Pressure Inputs		4-20 mA			Bridge		
	Pump Output	\$323.00	\$67.75		\$127.50	\$89.38	\$216.88
	Filter Output	\$323.00	\$67.75		\$127.50	\$89.38	\$216.88
	Section Output	\$323.00	\$67.75		\$127.50	\$89.38	\$216.88
Flow Rate Input		4-20 mA Flow Sensor					
		\$833.00	\$67.75		\$832.50	\$54.69	\$887.19
Discrete Inputs							
	On	\$40.00	\$30.69	\$70.69			
	Off	\$40.00	\$30.69	\$70.69			
	Heater On	\$40.00	\$30.69				
	Heater Off	\$40.00	\$30.69				
	Heater Overtemp		\$30.69	\$30.69			
	Section Overtemp		\$30.69	\$30.69			
	Reservoir level		\$30.69	\$30.69			
	Waterflow		\$30.69	\$30.69			
Discrete Outputs							
	Pump Contactor		\$62.81	\$62.81			
	Heater contactor		\$62.81	\$62.81			
	Heater PWM		\$97.81	\$97.81			
	Cost/Section Exchanger			\$950.81			\$1,716.56
	Cost/Magnet Exchanger			\$326.94			\$1,716.56
	Eight Section Heat Exchangers			\$7,606.50			\$13,732.50
	Three Magnet Heat Exchangers			\$980.81			\$5,149.69
	Ethernet adapter			\$914.00			
	Ethernet hub						
	EPICS Serial I/O						\$500.00
	PLC			\$503.00			\$635.00
	Battery Backup						\$1,000.00
	Electrical Materials			\$500.00			\$500.00
	Plumbing Materials			\$5,000.00			
	RTD Handheld Meter						
	Contingency			\$2,000.00			\$2,000.00
	Tax			\$875.22			\$1,175.86
	Total			\$18,379.53			\$24,693.05

5. SCHEDULE

Task		Time (days)	Who	Start	Complete
Modify Section Heat Exchanger Plumbing					
	Remove Heat Exchanger	5	Mark	4-Jul-2000	11-Jul-2000
	Remove 3 way valve	5	Mark	11-Jul-2000	18-Jul-2000
	Remove pneumatics	5	Mark	18-Jul-2000	25-Jul-2000
	Relocate heaters	5	Mark	25-Jul-2000	1-Aug-2000
	Repipe systems	10	Mark	1-Aug-2000	29-Aug-2000
	Vacation	10	Mark	4-Aug-2000	18-Aug-2000
	Install new diagnostic transducers	4	Mark	29-Aug-2000	3-Sep-2000
	Store reclaimed parts	3	Mark	3-Sep-2000	7-Sep-2000
	Refill systems	2	Mark	7-Sep-2000	10-Sep-2000
	Start System	1	Mark	10-Sep-2000	12-Sep-2000
Section Heat Exchanger Electrical Hook Up					
	Remove existing controls	8	Don, Wayne, Bob	4-Jul-2000	11-Jul-2000
	Wire transducers	14	Don, Wayne	11-Jul-2000	23-Jul-2000
	Wire controls and interlocks	24	Don, Wayne, Bob	23-Jul-2000	13-Aug-2000
	Programmable Controllers	7	Don, Wayne, Bob	13-Aug-2000	19-Aug-2000
	Test Control System and Interlocks	6	Don, Wayne	28-Sep-2000	4-Oct-2000
		6	Neil	28-Sep-2000	4-Oct-2000
Low Energy Heat Exchanger Plumbing					
	Disassemble existing system	2	Mark	12-Sep-2000	18-Sep-2000
	Vacation	4	Mark	5-Sep-2000	8-Sep-2000
	Reclaim parts/store unused components	2	Mark	18-Sep-2000	21-Sep-2000
	Build H/E skid	10	Mark	21-Sep-2000	5-Oct-2000
	Reconnect H/E	2	Mark	5-Oct-2000	8-Oct-2000
	Refill system/Start system	1	Mark	8-Oct-2000	9-Oct-2000
	Control system and interlocks	1	Mark	9-Oct-2000	11-Oct-2000
	Low energy system operational	1	Mark	11-Oct-2000	12-Oct-2000
Low Energy Heat Exchanger Electrical Hookup					
	Install electrical control box	2	Don, Wayne	19-Aug-2000	21-Aug-2000
	Wire pump heater, interlocks, control system	3	Don, Wayne	21-Aug-2000	23-Aug-2000
	Programmable Controller	1	Don, Wayne	23-Aug-2000	24-Aug-2000
	Test System	1	Don, Wayne	4-Oct-2000	4-Oct-2000
		1	Neil	4-Oct-2000	4-Oct-2000

Move Injection Extraction Heat Exchanger to ECS Heat Exchanger Mechanical					
	Disassembly Inj/Ext H/E	2	Mark	12-Oct-2000	15-Oct-2000
	Radiation Check all parts	1	Mark	15-Oct-2000	16-Oct-2000
	Reclaim parts / Store unused components	1	Mark	16-Oct-2000	18-Oct-2000
	Fabricate New ECS Heat Exchanger	3	Mark	18-Oct-2000	22-Oct-2000
	Disassemble ECS H/E	1	Mark	22-Oct-2000	23-Oct-2000
	Install New Heat Exchanger	2	Mark	23-Oct-2000	26-Oct-2000
	Refill system/start system	1	Mark	26-Oct-2000	28-Oct-2000
	Control system + interlocks	1	Mark	28-Oct-2000	29-Oct-2000
Move Injection Extraction Heat Exchanger to ECS Heat Exchanger Electrical					
	Install Electrical control boxes	2	Don, Wayne	24-Aug-2000	26-Aug-2000
	Wire control system	3	Don, Wayne	26-Aug-2000	29-Aug-2000
	Programmable Controller	1	Don, Wayne	29-Aug-2000	29-Aug-2000
	Test System	1	Don, Wayne	29-Oct-2000	30-Oct-2000
		1	Neil	29-Oct-2000	31-Oct-2000
Electrical Engineering/Documentation					
	Develop PLC schematic	5	Neil	4-Jul-2000	1-Aug-2000
	Develop Fieldpoint schematic	3	Neil	18-Jul-2000	3-Aug-2000
	Develop PLC Program	10	Neil	3-Aug-2000	28-Sep-2000
	CAD drawing of electrical schematic	5	Don	30-Oct-2000	8-Nov-2000
	Order PLC/Components	2	Neil	4-Jul-2000	2-Sep-2000
	Document PLC	5	Neil	30-Oct-2000	27-Nov-2000
Mechanical Engineering/Documentation					
	Section Heat Exchanger				
	Preliminary/detailed design	11	Mech Eng	28-Jul-2000	17-Aug-2000
	Order Parts	1	Mech Eng	17-Aug-2000	19-Aug-2000
	Upgrade Low Energy H/E to Plate-Plate				
	Conceptual Design	28	Mech Eng	19-Aug-2000	10-Oct-2000
	Preliminary/detailed design	13	Mech Eng	10-Oct-2000	3-Nov-2000
	Order Parts	1	Mech Eng	3-Nov-2000	5-Nov-2000
	Replace ECS Heat Exchanger				
	Conceptual Design	19	Mech Eng	5-Nov-2000	11-Dec-2000
	Preliminary/detailed design	14	Mech Eng	11-Dec-2000	6-Jan-2001
	Order Parts	1	Mech Eng	6-Jan-2001	8-Jan-2001